

Hyungjin Chung, Ph.D.

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Research interests Generative models, Alignment, Imaging, AI for medicine

Work Experience

Korea University	Seoul, Korea
(Incoming) Assistant Professor, Dept. of CSE	2026.09 –
EverEx	Seoul, Korea
Lead AI Research Scientist	2024.08 – Current
NVIDIA Research	San Jose, USA (remote)
Research Scientist Intern, AI4Science	2023.11 – 2024.01
Google Research	Mountain View, USA
Student Researcher, team LUMA	2023.07 – 2023.10
Los Alamos National Laboratory	Los Alamos, USA
Research intern, Applied math & Plasma physics	2022.06 – 2022.08

Education

KAIST	Daejeon, Korea
Ph.D., Bio & Brain Engineering	2019.03 – 2025.02
Advisor: Jong Chul Ye	
Thesis: <i>Practical approximations of posterior sampling in diffusion model-based inverse problems</i>	
Korea University	Seoul, Korea
B.S., Biomedical Engineering	2015.03 – 2019.02

Conf. publications *First authors, [†]Corresponding authors. If absent, first/last authors are first/corresponding.

[C23] ContrastiveCFG: Guiding Diffusion Sampling by Contrasting Positive and Negative Concepts
Jinho Chang, Changsun Lee, [Hyungjin Chung](#), Jong Chul Ye
ICML 2026

[C22] ReDirector: Robust Video Diffusion with Reversible Frame Reordering
Byeongjun Park, Byung-Hoon Kim, [Hyungjin Chung](#)[†], Jong Chul Ye[†]
CVPR 2026

[C21] Video Parallel Scaling: Aggregating Diverse Frame Subsets for VideoLLMs
[Hyungjin Chung](#), Hyelin Nam, Jiyeon Kim, Hyojun Go, Byeongjun Park, Junho Kim, Joonseok Lee, Seongsu Ha, Byung-Hoon Kim
CVPR 2026 Findings

[C20] A Foundational Brain Dynamics Model via Stochastic Optimal Control
Joonhyeong Park*, Byoungwoo Park*, Chang-Bae Bang, Jungwon Choi, [Hyungjin Chung](#), Byung-Hoon Kim[†], Juho Lee[†]
ICLR 2026

[C19] InvFusion: Bridging Supervised and Zero-shot Diffusion for Inverse Problems
Noam Elata*, [Hyungjin Chung](#)*, Jong Chul Ye, Tomer Michaeli, Michael Elad
NeurIPS 2025

[C18] Lesion-Aware Post-Training of Latent Diffusion Models for Synthesizing Diffusion MRI from CT Perfusion
Junhyeok Lee, Hyungwoong Kim, [Hyungjin Chung](#), Heeseong Eom, Jang Joon, Chul-Ho Sohn, Kyu Sung Choi
MICCAI 2025

- [C17] CapeLLM: Support-Free Category-Agnostic Pose Estimation with Multimodal Large Language Models
Junho Kim, [Hyungjin Chung](#)[†], Byung-Hoon Kim[†]
ICCV 2025
- [C16] VideoRFSplat: Direct Scene-Level Text-to-3D Gaussian Splatting Generation with Flexible Pose and Multi-View Joint Modeling
Hyojun Go*, Byeongjun Park*, Hyelin Nam, Byung-Hoon Kim, [Hyungjin Chung](#)[†], Changick Kim[†]
ICCV 2025
- [C15] SteerX: Creating Any Camera-Free 3D and 4D Scenes with Geometric Steering
Byeongjun Park*, Hyojun Go*, Hyelin Nam, Byung-Hoon Kim, [Hyungjin Chung](#)[†], Changick Kim[†]
ICCV 2025
- [C14] Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI
Won Jun Kim*, [Hyungjin Chung](#)*, Jemin Kim*, Byeongsu Sim, Sangmin Lee, Jong Chul Ye
CVPR 2025
- [C13] CFG++: Manifold-constrained Classifier Free Guidance for Diffusion Models
[Hyungjin Chung](#)*, Jeongsol Kim*, Geon-Yeong Park*, Hyelin Nam*, Jong Chul Ye
ICLR 2025
- [C12] Regularization by texts for latent diffusion inverse solvers
Jeongsol Kim*, Geon-Yeong Park*, [Hyungjin Chung](#), Jong Chul Ye
ICLR 2025 (spotlight)
- [C11] Deep Diffusion Image Prior for Efficient OOD Adaptation in 3D Inverse Problems
[Hyungjin Chung](#) and Jong Chul Ye
ECCV 2024
- [C10] Prompt-tuning Latent Diffusion Models for Inverse Problems
[Hyungjin Chung](#), Jong Chul Ye, Peyman Milanfar, Mauricio Delbracio
ICML 2024
- [C9] Decomposed Diffusion Sampler for Accelerating Large-Scale Inverse Problems
[Hyungjin Chung](#), Suhyeon Lee, Jong Chul Ye
ICLR 2024
- [C8] Direct Diffusion Bridge using Data Consistency for Inverse Problems
[Hyungjin Chung](#), Jeongsol Kim, Jong Chul Ye
NeurIPS 2023
- [C7] Improving 3D Imaging with Pre-Trained Perpendicular 2D Diffusion Models
Suhyeon Lee*, [Hyungjin Chung](#)*, Minyoung Park, Jonghyuk Park, Wi-Sun Ryu, Jong Chul Ye
ICCV 2023
- [C6] Score-based Diffusion Models for Bayesian Image Reconstruction
Michael T. Mccann, [Hyungjin Chung](#), Jong Chul Ye, Marc L. Klasky
ICIP 2023
- [C5] Parallel Diffusion Models of Operator and Image for Blind Inverse Problems
[Hyungjin Chung](#)*, Jeongsol Kim*, Sehui Kim, Jong Chul Ye
CVPR 2023
- [C4] Solving 3d inverse problems using pre-trained 2d diffusion models
[Hyungjin Chung](#)*, Dohoon Ryu*, Michael T. Mccann, Marc L. Klasky, Jong Chul Ye
CVPR 2023
- [C3] Diffusion Posterior Sampling for General Noisy Inverse Problems
[Hyungjin Chung](#)*, Jeongsol Kim*, Michael T. Mccann, Marc L. Klasky, Jong Chul Ye
ICLR 2023 (Notable-top-25%)

[C2] Improving Diffusion Models for Inverse Problems using Manifold Constraints
[Hyungjin Chung*](#), Byeongsu Sim*, Dohoon Ryu, Jong Chul Ye
NeurIPS 2022

[C1] Come-Closer-Diffuse-Faster: Accelerating Conditional Diffusion Models for Inverse Problems through Stochastic Contraction
[Hyungjin Chung](#), Byeongsu Sim, and Jong Chul Ye
CVPR 2022

Journal publications

[J15] Deep Learning for Deep Learning Performance: How Much Data Is Needed in Biomedical Imaging?

Kyu Sung Choi, Junhyeok Lee, [Hyungjin Chung](#), Jeong-Hoon Lee
PLOS One, 2026

[J14] Label-independent Framework for Objective Evaluation of Cosmetic Outcome in Breast Cancer

Sangjoon Park, Yong Bae Kim, Jee Suk Chang, Seo Hee Choi, [Hyungjin Chung](#), Ik Jae Lee, Hwa Kyung Byun
Artificial Intelligence in Medicine, 2025

[J13] Steerable Conditional Diffusion for Out-of-Distribution Adaptation in Medical Image Reconstruction

Alexander Denker*, Riccardo Barbano*, [Hyungjin Chung*](#), Tae Hoon Roh, Simon Arrdige, Peter Maass, Bangti Jin, Jong Chul Ye
IEEE TMI, 2025

[P2] Objective and Interpretable Breast Cosmesis Evaluation with Attention Guided Denoising Diffusion Anomaly Detection Model

Sangjoon Park, Yong Bae Kim, Jee Suk Chang, Seo Hee Choi, [Hyungjin Chung](#), Ik Jae Lee, Hwa Kyung Byun
IJROBP, 2024

[J12] Fundus image enhancement through direct diffusion bridges

Sehui Kim*, [Hyungjin Chung*](#), Se Hie Park, Eui-Sang Chung, Kayoung Yi, Jong Chul Ye
IEEE JBHI, 2024

[J11] MR Image Denoising and Super-Resolution Using Regularized Reverse Diffusion

[Hyungjin Chung](#), Eun Sun Lee, Jong Chul Ye
IEEE TMI, 2022

[J10] Low-dose sparse-view HAADF-STEM-EDX tomography of nanocrystals using unsupervised deep learning

Eunju Cha*, [Hyungjin Chung*](#), Jaeduck Jang, Junho Lee, Eunha Lee, Jong Chul Ye
ACS Nano, 2022

[J9] Score-based diffusion models for accelerated MRI

[Hyungjin Chung](#) and Jong Chul Ye
Medical Image Analysis, 2021

[J8] Unsupervised Deep Learning Methods for Biological Image Reconstruction and Enhancement

Mehmet Akçakaya, Burhaneddin Yaman, [Hyungjin Chung](#), Jong Chul Ye
IEEE SPM, 2021

[J7] A Deep Learning Model for Diagnosing Gastric Mucosal Lesions Using Endoscopic Images: Development, Validation, and Method Comparison

Joon Yeul Nam*, [Hyungjin Chung*](#), Kyu Sung Choi*, Hyuk Lee* et al.
Gastrointestinal Endoscopy, 2021

[J6] Feature Disentanglement in generating three-dimensional structure from two-dimensional slice with sliceGAN

Hyungjin Chung, Jong Chul Ye
Nature Machine Intelligence, 2021

[J5] Missing Cone Artifacts Removal in ODT using Unsupervised Deep Learning in Projection Domain
 Hyungjin Chung*, Jaeyoung Huh*, Geon Kim, Yong Keun Park, Jong Chul Ye
IEEE TCI, 2021

[J4] Two-Stage Deep Learning for Accelerated 3D Time-of-Flight MRA without Matched Training Data
 Hyungjin Chung, Eunju Cha, Leonard Sunwoo, Jong Chul Ye
Medical Image Analysis, 2021

[J3] Deep learning STEM-EDX tomography of nanocrystals
 Yoseob Han*, Jaeduck Jang*, Eunju Cha*, Junho Lee*, Hyungjin Chung* et al.
Nature Machine Intelligence, 2021 (March Issue cover)

[J2] Unpaired training of deep learning tMRA for flexible spatio-temporal resolution
 Eunju Cha, Hyungjin Chung, Eung Yeop Kim, Jong Chul Ye
IEEE TMI, 2021

[J1] Unpaired deep learning for accelerated MRI using optimal transport driven cycleGAN
 Gyutaek Oh, Byeongsu Sim, Hyungjin Chung, Leonard Sunwoo, Jong Chul Ye
IEEE TCI, 2020

Books

[B2] Generative Machine Learning Models in Medical Image Computing
 Chapter 7: Diffusion Models for Inverse Problems in Medical Imaging
 Hyungjin Chung, Jong Chul Ye

[B1] Deep Learning for Biomedical Image Reconstruction
 Chapter 12: Image Synthesis in Multi-Contrast MRI with Generative Adversarial Networks
 Tolga Çukur, Mahmut Yurt, Salman Ul Hassan Dar, Hyungjin Chung, Jong Chul Ye

Awards

31st Samsung Humantech Silver Award	2025.02
Google Conference Scholarship	2024.05
30th Samsung Humantech Gold Award	2024.02
29th Samsung Humantech Gold Award	2023.02
2020-2024 BISPL Best Researcher Award	2020-2024.12

Professional Service

Committee Member	
KSIAM Imaging Science	2026-2027
Technical Group on Image Processing, IEIE	2026-
Grant Reviewer	
The Royal Society (Leverhulme Trust Senior Fellowship)	2026
Reviewer (Conference)	
ICML (<i>Gold reviewer</i> 2026)	2024-
ICLR	2024-
NeurIPS	2022-
NeurIPS Datasets&Benchmarks	2023-
CVPR	2023-
ECCV	2022-
ICCV	2023-
AAAI	2025-
SIGGRAPH	2026-
SIGGRAPH Asia	2025-

MICCAI 2022-
Reviewer (Journal)
 NEJM AI
 Nature Communications
 npj Digital Medicine
 Medical Image Analysis
 IEEE TMI (*Gold Distinguished reviewer 2024, Bronze Distinguished reviewer 2023*)
 IEEE TPAMI, TCI, TSP, TIP, SPS, SPL, TRPMS
[See full list](#)

Invited talks

Physically Plausible Generative AI
 - KAIST AI 2026.03
 - DGIST AI 2026.03
 - ETH-PRS 2026.02
 - Joint Winter School on Image Understanding and Image Processing by IEIE 2026.01
Towards motion understanding and generation in videos
 - Dept. of Applied Statistics and Data Science, Yonsei University 2025.10
 - Korean Society of Digital Health 2025.09
Texts in inverse problem solving using diffusion models
 - University of Michigan 2024.10
Tutorial on Denoising Diffusion Model: Fundamentals & Applications
 - IEIE: Winter School on Biomedical Signal Processing 2024.02
Adapting diffusion models for inverse problems
 - UCLA, Caltech: Grundfest Memorial Lecture Series in Graphics and Imaging 2024.02
 - 2023 NeurIPS Workshop on diffusion models 2023.12
 - Google Research 2023.10
Advances in diffusion models and their applications to inverse problems
 - Guest Lecture, Korea University 2023.11
Generative (diffusion) models for medical imaging
 - KoSAIM 2025 summer school 2025.08
 - International Congress on Magnetic Resonance Imaging (ICMRI) 2023 2023.11
 - Michigan State University 2023.09
 - Stanford MedAI 2023.08
 - MGH, School of Medicine, Harvard University 2023.08
 - BRIC academic webinar 2023.03
 - 45th meeting, The Korean Society of Abdominal Radiology, 2022 2022.06
Diffusion models: foundations and applications in biomedical imaging
 - IEEE International Symposium on Biomedical Imaging (ISBI) 2023 2023.05
Diffusion models for inverse problems
 - The Korean Institute of Broadcast and Media Engineers 2025.12
 - University of Minnesota 2025.11
 - LANL 2024.11
 - IPA seminar, Korea University 2024.09
 - Krafton AI 2024.09
 - DRGem 2024.08
 - LG AI Research 2024.08
 - Twelve Labs 2024.06
 - AI SEOUL 2024 2024.02
 - Inference & control group seminar, Donders Institute, Radboud Univ. 2023.01
 - LANL T-CNLS seminar, 2022 2022.08

Preprints

[P7] Advancing Ultra Low-Field MRI with Synthetic Data and Deep Learning-Based Image Enhancement for Brain Volume Analysis

Peter Hsu, Elisa Marchetto, [Hyungjin Chung](#), Dohun Lee, Jong Chul Ye, Daniel Sodickson, Jelle Veraart, Patricia Johnson

[P6] ContextMRI: Enhancing Compressed Sensing MRI through Metadata Conditioning

[Hyungjin Chung](#)^{*}, Dohun Lee^{*}, Zihui Wu, Byung-Hoon Kim, Katie Bouman, Jong Chul Ye

[P4] ACDC: Autoregressive coherent multimodal generation using diffusion correction

[Hyungjin Chung](#)^{*}, Dohun Lee^{*}, Jong Chul Ye

[P3] A survey on diffusion models for inverse problems

Giannis Daras, [Hyungjin Chung](#), Chieh-Hsin Lai, Yuki Mitsufuji, Jong Chul Ye, Peyman Milanfar, Alexandros G Dimakis, Mauricio Delbracio

[P2] Amortized Posterior Sampling with Diffusion Prior Distillation

Abbas Mammadov^{*}, [Hyungjin Chung](#)^{*}, Jong Chul Ye

[P1] Generative AI for Medical Imaging: extending the MONAI Framework

Pinaya *et al.* ([Hyungjin Chung](#): Contributing author)

Patent

US patent application

- Score-based Diffusion Model for Accelerated MRI and Apparatus thereof 2023

Korea patent publication

- Crowd Deep Learning Method of Medical Artificial Intelligence and Apparatus thereof 2025

- Score-based Diffusion Model for Accelerated MRI and Apparatus thereof 2024

- Task-agnostic image processing method and apparatus using transformer and federated split learning 2024

- Tomography image processing method using neural network based on unsupervised learning to remove missing cone artifacts and apparatus therefor 2023

- Two-Stage unsupervised learning method for 3D Time-of-flight MRA reconstruction and the apparatus thereof 2023

Korea patent application

- Accelerating method of conditional diffusion models for inverse problems using stochastic contraction and the apparatus thereof 2021

- Extreme condition reconstruction method HAADF-STEM-EDX tomography using unsupervised deep learning and the apparatus thereof 2021

References

Byung-Hoon Kim	2024.08 — Current
CIO (EverEx)	egyptdj@yonsei.ac.kr
Jong Chul Ye	2019.03 — 2025.02
Ph.D. advisor (KAIST)	jong.ye@kaist.ac.kr
Mauricio Delbracio	2023.07 — 2023.11
Host (Google)	mdelbra@google.com
Peyman Milanfar	2023.07 — 2023.11
Host (Google)	milanfar@google.com
Michael T. McCann	2022.06 — 2022.08
Host (LANL)	mccann@lanl.gov